

Lecture ٦.٢

Inefficiency of Random Access-Based Multiple Access Protocols

١. Overview

Medium Access Control (MAC) protocols determine how devices share a common communication medium. Among them, random access-based MAC protocols—such as ALOHA, CSMA, and IEEE ٨٠٢.١١ (WiFi)—enable decentralized coordination using randomized backoff mechanisms. While this decentralized approach avoids centralized control overhead, it suffers from fundamental inefficiencies, particularly under moderate to high network load, due to repeated collisions and idle channel times.

٢. The Backoff Mechanism and Its Limitations

In CSMA/CA based MAC, each node selects a random number from a contention window (CW) and begins a countdown. The first node to reach zero transmits, while others wait. Although conceptually simple, this process leads to significant inefficiencies:

- **Idle Channel Time:** During the countdown, the channel remains idle, even though no transmission occurs. This results in underutilization of available bandwidth.
- **Collision Probability:** When two or more nodes pick the same backoff value, simultaneous transmission leads to packet collisions. These collisions cause retransmissions and an exponential increase in CW, further delaying future transmissions.

- In IEEE ٨٠٢.١١, with $CW=١٦$, even with ١٦ nodes, the probability of a collision is $\sim ١٨\%$.
- Studies (مراجع) indicate over ٣٠% throughput reduction in congested wireless LANs due to excessive backing off.
- At higher data rates (e.g., ٥٤ Mbps), shorter packet channel times exacerbate inefficiency since fixed slot durations (e.g., ٩ μs) consume a greater fraction of the channel time.

٣. Detailed Inefficiencies

٣.١ Wasted Channel Time from Idle Slots

- Slot duration is lower bounded by propagation delay, clear channel assessment (CCA) time, and processing delays. For ٨٠٢.١١, this is typically $\sim ٢٢٢ \mu s$.
- With high bitrate transmissions, e.g., ٥٤ Mbps, the channel time is only $\sim ٢٢٢ \mu s$. If backoff involves, say, ٨ slots on average, the idle time becomes $\sim ١.٧8 ms$, which is nearly ٢٥% of total channel time wasted before actual transmission.

٣.٢ Exponential Backoff Under Congestion

- In IEEE ٨٠٢.١١, Binary Exponential Backoff (BEB) increases CW size as:
 $CW = 2^{\text{number of failed attempts}} \times CW_{min} - ١$, for each failed attempt (up to $CW_{max} = ١٠٢٤$).
- At ١٠ active nodes, probability of at least two nodes choosing the same slot exceeds drastically, reducing throughput and increasing latency.

- Simulation and analytical results from [10] show that throughput saturates as network density increases beyond a threshold.

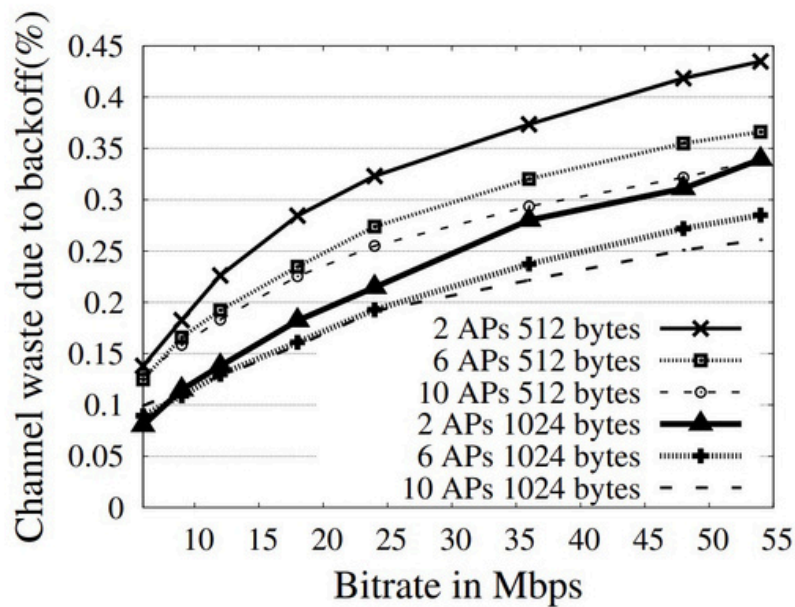


Fig. 1. Overhead of 802.11 backoff. Larger fraction of channel wasted with smaller packets at higher bitrates.

Figure 1 illustrates channel under-utilization across varying data rates and number of contending nodes. As node density increases, idle time and collision probability increase disproportionately, demonstrating a non-linear degradation of performance.

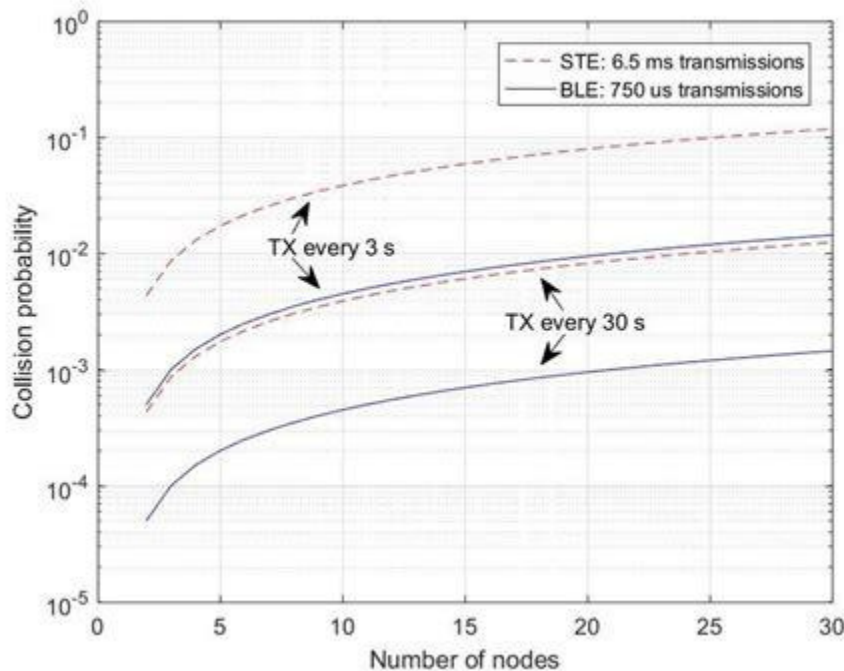


Figure 2: Collision Probability vs. Number of Nodes

This figure shows how collision probability increases nonlinearly with the number of contending nodes. Even with 10 nodes, collisions exceed 10%, leading to frequent retransmissions and reduced throughput. The inefficiency grows sharply with scale, limiting the practicality of random access MAC in dense networks.

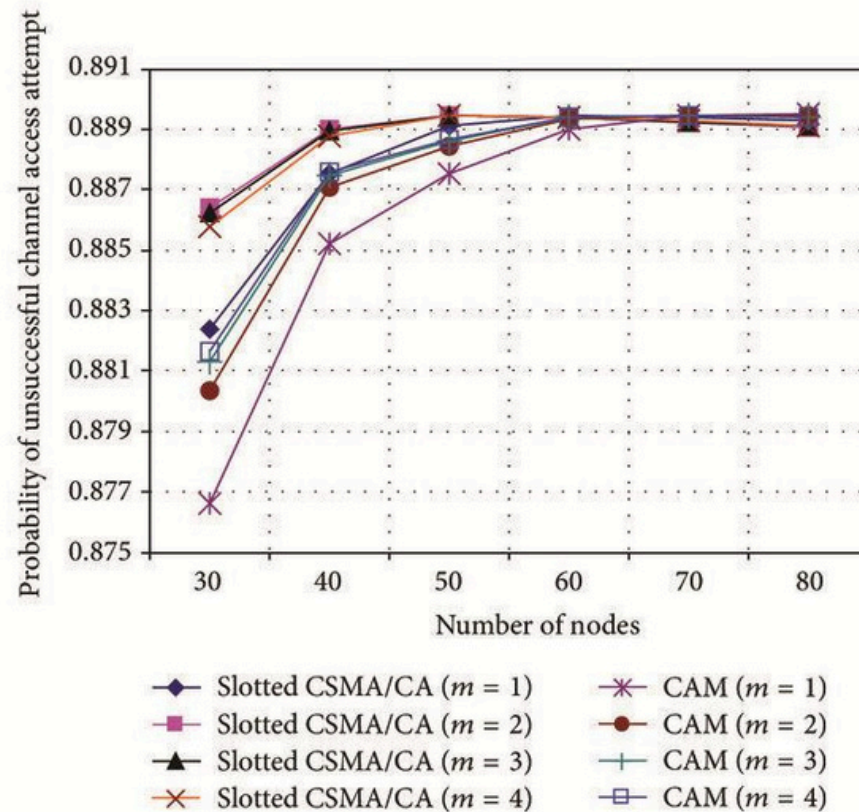


Figure 1: Collision Probability vs. Channel Load

As channel load increases, collision probability rises rapidly exceeding beyond a threshold (e.g., 1 pkt/sec). This illustrates how CSMA/CA becomes highly inefficient under high traffic, highlighting the need for smarter or hybrid MAC strategies.

o. Implications and Future Directions

The drawbacks of random-access MAC highlight the need for:

- Smarter contention resolution, e.g., using machine learning or adaptive CW adjustment.
- Hybrid protocols, combining scheduled and contention-based approaches.
- Resource reservation mechanisms, such as Semi-Random Backoff (الرجوع العشوائي), which aim to reduce collision probability.

References

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